ELEC 872 Week 5 Lab

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1 Experimental Setup and Reference Model

All experiments use the OPPORTUNITY dataset with the gesture task. The input is 113 channels from body-worn accelerometers, gyroscopes and magnetometers sampled at 30 Hz, and each window is labelled as one of 17 gestures or the Null class (no gesture). Training uses all five daily-living runs and the drill run of subject 1, and daily-living runs 1 and 2 and the drill run of subjects 2 and 3. Validation uses daily-living run 3 of subjects 2 and 3, and testing uses daily-living runs 4 and 5 of the same two subjects. Windows are 24 samples long (0.8 s) and move by 12 samples, which gives 41,399 training, 5,076 validation and 9,890 test windows. Of the test windows, 8,235 (83%) are Null.

The reference model follows the paper: four convolutional layers with 64 filters of length 5, two LSTM layers with 128 units, and dropout of 0.5, for 3,965,778 parameters. It is trained with RMSProp (learning rate 0.001, decay 0.9) on batches of 100 windows for up to 100 epochs. Training stops when the validation weighted F1 has not improved for 20 epochs, and the epoch with the best validation weighted F1 is evaluated once on the test windows.

We report two scores. Weighted F1 averages the per-class F1 with weights proportional to the number of test windows of each class, so the Null class dominates it. Macro F1 is the plain average over the 18 classes, so every gesture counts equally. Each configuration in Sections 1, 5, 6 and 7 was trained three times with different random seeds, and we report the mean and standard deviation over the three runs. These runs used one slice (1g.10gb) of an NVIDIA H100 GPU and took 6 to 13 minutes each.

The reference model reaches a test weighted F1 of 0.891 ± 0.004, a macro F1 of 0.580 ± 0.029 and an accuracy of 0.893. The paper reports a weighted F1 of 0.915 for the same task, so our reproduction is 0.024 lower. The single Colab run used as the baseline in the LSTM section gave 0.886, slightly below the lowest of our three runs (0.887). The macro F1 varies by 0.029 between seeds of the same configuration. In the following sections we therefore do not treat a difference smaller than about 0.03 in macro F1 as an effect of the change being tested.

2 Influence of LSTM on Model Performance

To test the paper's claim that the LSTM is necessary to capture the temporal dimension of the signal data, we train and validate two variations of the model:

- A no RNN version where the LSTM is replaced with two dense layers
- A GRU variant in which LSTM cells are replaced with GRU

	Accuracy	F1	Avg Precision	Avg Recall
Baseline	0.890	0.886	0.887	0.890
No RNN	0.874	0.857	0.860	0.874
GRU-Variant	0.896	0.896	0.901	0.896

3 Influence of CNN on Model's Performance

To test the CNN's influence on model success, the number of convolution layers was varied, and performance measured:

	Accuracy	F1	Avg Precision	Avg Recall
Baseline	0.890	0.886	0.887	0.890
Number CNNs = 0	0.854	0.806	0.789	0.854
Number CNNs = 2	0.893	0.893	0.896	0.893

4 How long should the window be?

5 How Much Regularization Is Needed?

The paper uses dropout of 0.5 and no other regularization, and states in Section 5.4 that higher dropout made no difference, without giving numbers. We test whether dropout, weight decay and data augmentation change the test results.

Because the reference model already has no weight decay and no augmentation, removing either from it would leave it unchanged. We therefore switched on all three regularizers and removed one at a time. Dropout of 0.5 is applied to the convolutional features, between the two LSTM layers and before the output layer. Weight decay adds an L2 penalty of 10^{-4} to the RMSProp update. Augmentation multiplies every channel of every training batch by a random gain drawn from $\mathcal{N}(1, 0.1^2)$ and adds Gaussian noise with a standard deviation of 0.01 (the data are normalized to $[0, 1]$). To measure over-fitting, we also scored each saved model on the training windows with dropout and augmentation switched off, and compared its macro F1 there with the test macro F1.

	Test		Training	
Setting	Weighted F1	Macro F1	macro F1	Gap
Reference (dropout only)	0.891 ± 0.004	0.580 ± 0.029	0.902	0.32
All three regularizers	0.883 ± 0.007	0.562 ± 0.012	0.877	0.32
No dropout	0.885 ± 0.004	0.569 ± 0.013	0.940	0.37
No weight decay	0.881 ± 0.005	0.553 ± 0.041	0.852	0.30
No augmentation	0.890 ± 0.006	0.578 ± 0.016	0.919	0.34

Table 1: Regularization settings, mean ± standard deviation over three seeds. All three regularizers means dropout, weight decay and augmentation together; the last three rows each remove one of them. Gap is the training macro F1 minus the test macro F1.

Table 1 shows that the test macro F1 of all five settings lies between 0.553 and 0.580. The largest difference, 0.027, is smaller than the seed-to-seed variation of the reference model (0.029). Adding weight decay and augmentation on top of dropout did not improve the results: the weighted F1 was 0.883 against 0.891 for dropout alone. Removing dropout raised the training macro F1 from 0.877 to 0.940 and widened the gap between training and test from 0.32 to 0.37 (Figure 1), while the test macro F1 stayed at 0.569 against 0.562.

Dropout therefore reduces over-fitting, but in this setting that does not change the test performance, and weight decay and augmentation have no measurable effect either. This agrees with the paper's statement that the regularization settings matter less than the architecture. With three seeds per setting, we cannot rule out effects smaller than about 0.03 in macro F1.

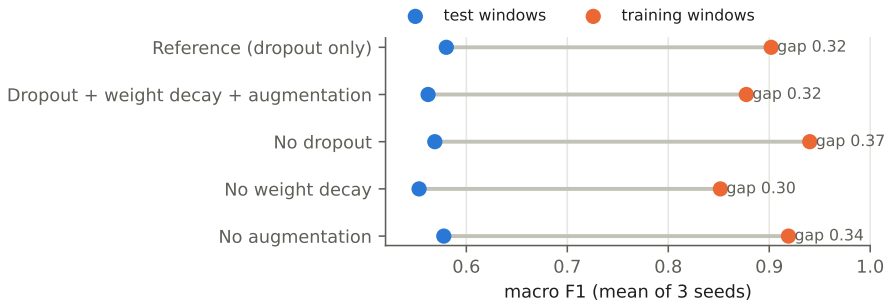


Figure 1: Macro F1 on the training windows (orange) and on the test windows (blue) for each regularization setting, mean of three seeds. The label gives the difference.

6 Does a Learning-Rate Warm-up Make Training More Stable?

The paper trains with RMSProp at a fixed learning rate of 0.001 and reports that it gave the best results, without giving numbers. We tested whether a learning-rate warm-up makes training more stable. The learning rate rises linearly from almost zero to 0.001 over the first 5 epochs and is updated after every mini-batch, which is 2,070 updates at 414 mini-batches per epoch. After the warm-up the training is identical to the reference.

We measured stability in three ways. The early loss variation is the standard deviation of each mini-batch loss relative to the mean of the surrounding 50 mini-batches, over epochs 1 to 5. The largest validation drop is the largest fall of the validation weighted F1 from one epoch to the next. The validation-loss increases count the epochs among the first 15 in which the validation loss went up.

	Reference	Warm-up (5 epochs)
Test weighted F1	0.891 ± 0.004	0.894 ± 0.004
Test macro F1	0.580 ± 0.029	0.604 ± 0.016
Early loss variation (epochs 1 to 5)	0.209 ± 0.004	0.188 ± 0.008
Largest validation weighted F1 drop	0.032 ± 0.019	0.021 ± 0.011
Validation-loss increases (first 15 epochs)	6.0 ± 1.0	6.7 ± 1.5
Best epoch	35 (21, 62, 21)	56 (36, 62, 71)

Table 2: Learning-rate warm-up, mean $\pm$ standard deviation over three seeds. Lower values of the three stability measures mean more stable training. The best epoch is the mean, with the three seeds in brackets.

With the warm-up, the early loss variation fell from 0.209 to 0.188 (Table 2), and the difference is larger than the spread between seeds. The largest validation drop fell from 0.032 to 0.021, but the seeds overlap, and the number of validation-loss increases did not change. Figure 2 shows one seed. During the warm-up the training loss falls more slowly, and the reference run shows a single sharp drop of the validation weighted F1 to 0.82 at epoch 38 that the warm-up run does not have. The warm-up also delayed convergence, as the best epoch moved from 35 to 56 on average.

The warm-up makes the first epochs somewhat smoother, but it does not give a clearly better model. The test macro F1 rose by 0.024 to 0.604, which is within the seed-to-seed variation of the reference. RMSProp at 0.001 already trains this model without serious instability.

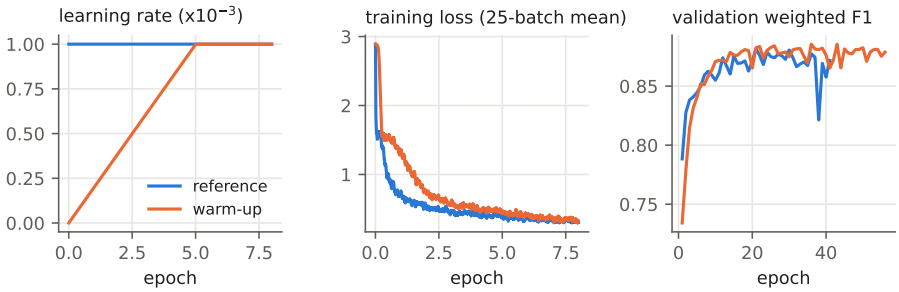


Figure 2: Learning rate (left) and training loss averaged over 25 mini-batches (middle) for the first 8 epochs, and validation weighted F1 over the whole run (right), for the reference (blue) and the warm-up (orange). One seed.

7 Does Handling Class Imbalance Help Rare Gestures?

The test set is dominated by the Null class, and several gestures have fewer than 100 test windows. The paper does not address this imbalance and reports only weighted F1. We replaced the cross-entropy loss with the Focal Loss of Lin et al. (2017),

$$\mathrm{FL}(p_t) = -(1 - p_t)^{\gamma} \log p_t, \tag{1}$$

where p_t is the predicted probability of the true class. The factor $(1 - p_t)^{\gamma}$ reduces the loss of windows the model already classifies confidently, which in this dataset are mostly Null windows. We used $\gamma = 2$, the value Lin et al. found best, and no class weights, so only the shape of the loss changes. The best epoch is still chosen on the validation weighted F1.

	Paper	Cross-entropy	Focal Loss ($\gamma = 2$)
Test weighted F1	0.915	0.891 ± 0.004	0.892 ± 0.003
Test macro F1	not reported	0.580 ± 0.029	0.589 ± 0.029
Macro F1, 17 gestures	not reported	0.558	0.567
Null precision	not reported	0.942	0.947
Null recall	not reported	0.958	0.949

Table 3: Cross-entropy (the reference model) and Focal Loss, mean over three seeds ($\pm$ standard deviation where shown). The paper reports only the weighted F1.

The Focal Loss did not change the overall scores (Table 3). The weighted F1 was 0.892 against 0.891, and the macro F1 rose by 0.009, far less than the seed-to-seed variation of 0.029. Figure 3 shows the per-gesture F1. Nine of the 17 gestures improved, and over the 12 gestures with fewer than 100 test windows

the mean change was 0.000. The drawer gestures, which are the rarest, moved in both directions: Close Drawer 1 and Open Drawer 2 improved by 0.07, while Close Drawer 2 and Open Drawer 1 dropped by 0.04. The Null recall fell from 0.958 to 0.949, which means the model predicted gestures slightly more often. Neither loss closes the gap of about 0.02 to the paper's weighted F1.

Focal Loss with $\gamma = 2$ alone does not help the rare gestures in this setting, and the drawer gestures remain between 0.31 and 0.57 F1. Focal Loss down-weights windows by how confidently they are classified, not by how rare their class is. Adding the class weights (α) of Lin et al. or sampling the rare gestures more often would be the next step to test.

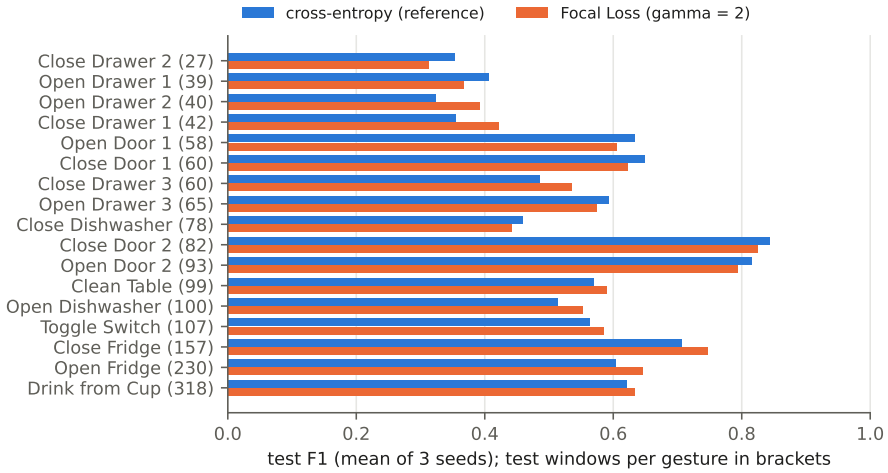


Figure 3: Test F1 of each gesture with cross-entropy (blue) and Focal Loss (orange), mean of three seeds, rarest gesture first. The number of test windows of each gesture is in brackets.